# FSM-Based Priority Traffic Light Controller

Hardware Architecture, RTL Logic & Behavioral Verification in Verilog-2001

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|---|---|
| **Target Architecture:** Xilinx 7-Series / Artix-7 / Zynq | **Hardware Language:** Verilog-2001 (Synthesizable) |
| **Simulation Tool:** AMD Xilinx Vivado (XSim) | **Author / Project:** joeljo777 / ASICD_Project |
| **Intersection Model:** 4-Way (North-South vs East-West) | **Priority Scheme:** Two-Tier Emergency Preemption |

## 1. Executive Summary & Design Philosophy

Modern urban traffic junctions require reliable digital control logic capable of balancing high-throughput normal vehicular flow with immediate, collision-free preemption for high-priority emergency vehicles (ambulances, fire engines, police units). This project develops a fully synthesizable **Moore Finite State Machine (FSM)** in Verilog-2001, specifically engineered for Xilinx FPGAs and verified using AMD Vivado Behavioral Simulation.

**Primary Architectural Objectives:**

- **Deterministic Anti-Collision Guarantee:** Conflicting traffic lanes can never simultaneously display Green or Yellow signals. Continuous hardware assertions verify this invariant at every clock edge.
- **Braking Safety Enforcement:** Active Green lights are never snapped abruptly to Red upon emergency detection. Vehicles traveling at speed require bounded deceleration time (Yellow) and intersection clearance (All-Red) to eliminate lateral T-bone collisions.
- **Two-Tier Dynamic Preemption:** North-South (Arterial Main Road) holds primary emergency priority over East-West (Side Street). Simultaneous emergency alerts are arbitrated deterministically without deadlock.
- **Glitch-Free Output Registers:** Output light signals are registered on clock edges, eliminating combinatorial hazards on physical FPGA output pins.

***CRITICAL SAFETY INTERLOCK:*** *Under all preemption circumstances (including unexpected emergency vehicle arrivals, de-assertions, and system resets), the controller enforces full legal Yellow duration (3s) and All-Red clearance (1s) before switching right-of-way. This mimics real-world Department of Transportation (DOT) safety standards.*

**I/O Port Specifications:**

| Signal | Direction | Width | Active Level | Functional Role |
|---|---|---|---|---|
| `clk` | Input | 1 bit | Rising Edge | Master system oscillator (e.g., 100 MHz on Basys 3) |
| `rst_n` | Input | 1 bit | Active-Low | Asynchronous master system reset (initializes to S_NS_GREEN) |
| `emerg_ns` | Input | 1 bit | Active-High | Optical / acoustic sensor for North-South emergency vehicle |
| `emerg_ew` | Input | 1 bit | Active-High | Optical / acoustic sensor for East-West emergency vehicle |
| `ns_lights` | Output | 3 bits | Active-High | North-South lamp bus: [2]=Red, [1]=Yellow, [0]=Green |
| `ew_lights` | Output | 3 bits | Active-High | East-West lamp bus: [2]=Red, [1]=Yellow, [0]=Green |
| `state_out` | Output | 3 bits | Binary Code | Current FSM state index for debug / 7-segment display |

## 2. Finite State Machine (FSM) State Architecture

The controller is formulated as a 6-state synchronous Moore FSM. Each state corresponds to a mutually exclusive traffic signalling phase with defined durations and preemption branch policies:

| State | Encoding | NS {R,Y,G} | EW {R,Y,G} | Time (s) | Operational Description & Emergency Preemption |
|---|---|---|---|---|---|
| **S_NS_GREEN** | `3'b000` | **GREEN (001)** | **RED (100)** | 10 s | Main Street traffic active. If emerg_ns asserts, green is held indefinitely. If emerg_ew asserts, early transition to S_NS_YELLOW is initiated. |
| **S_NS_YELLOW** | `3'b001` | **YELLOW (010)** | **RED (100)** | 3 s | NS traffic braking buffer. Must run full 3s without interruption to permit safe vehicular stopping. |
| **S_ALL_RED_1** | `3'b010` | **RED (100)** | **RED (100)** | 1 s | All-Red clearance. Once elapsed: if emerg_ns is active, loops straight back to S_NS_GREEN; otherwise advances to S_EW_GREEN. |
| **S_EW_GREEN** | `3'b011` | **RED (100)** | **GREEN (001)** | 7 s | Side Street traffic active. If emerg_ew asserts, green is held. If emerg_ns asserts, green is immediately truncated and routed through S_EW_YELLOW. |
| **S_EW_YELLOW** | `3'b100` | **RED (100)** | **YELLOW (010)** | 3 s | EW traffic braking buffer. Runs full 3s safety interval without preemption truncation. |
| **S_ALL_RED_2** | `3'b101` | **RED (100)** | **RED (100)** | 1 s | All-Red clearance. Once elapsed: if emerg_ew && !emerg_ns, loops to S_EW_GREEN; otherwise returns to S_NS_GREEN. |

### Priority Arbitration Rules & Conflict Handling:

1. **Single Emergency on Opposing Road:** If an emergency vehicle triggers while the opposite road is green, the green timer is cancelled immediately and the current road enters its Yellow phase. Once the yellow and all-red clearance intervals conclude, the emergency road receives Green.

2. **Extended Green Duration:** If an emergency vehicle remains in transit across the intersection, the active green state is maintained indefinitely until the sensor signal drops back to zero.

3. **Simultaneous Emergency Arbitration:** If both emerg_ns and emerg_ew are asserted simultaneously (e.g., competing emergency calls):

- North-South (Arterial Road) is serviced first due to higher traffic capacity and safety classification.
- East-West is latched and granted green immediately after North-South clears its emergency and completes yellow/red clearance.
- This eliminates priority inversion and prevents system starvation.

# 3. Verilog RTL Implementation Analysis

The complete synthesizable design resides in traffic_light_controller.v. It follows industry-standard two-process FSM coding style with dedicated prescaling and registered output buffers:

## 3.1 1-Hz Tick Generator (Hardware Prescaler)

Directly counting FPGA clock cycles (100 MHz = 100,000,000 cycles/sec) in FSM comparison logic requires wide counters and adds logic depth to timing paths. Instead, a dedicated prescaler generates a single-cycle strobe tick_1s:

```
always @(posedge clk or negedge rst_n) begin
    if (!rst_n) begin
        clk_div_cnt <= 32'd0;
        tick_1s     <= 1'b0;
    end else begin
        if (clk_div_cnt >= CLK_FREQ_HZ - 1) begin
            clk_div_cnt <= 32'd0;
            tick_1s     <= 1'b1;  // Single-cycle enable strobe
        end else begin
            clk_div_cnt <= clk_div_cnt + 1'b1;
            tick_1s     <= 1'b0;
        end
    end
end
```

## 3.2 State Timer with Auto-Reset and Anti-Saturation

The state duration counter sec_cnt increments only when tick_1s pulses. It automatically resets to zero on every state entry (state_changed). An upper bound saturation clamp prevents rollover if an emergency vehicle extends green beyond 63 seconds:

```
assign state_changed = (current_state != next_state);

always @(posedge clk or negedge rst_n) begin
    if (!rst_n) begin
        sec_cnt <= 6'd0;
    end else if (state_changed) begin
        sec_cnt <= 6'd0;  // Auto-reset upon state transition
    end else if (tick_1s) begin
        if (sec_cnt < 6'd63)
            sec_cnt <= sec_cnt + 1'b1; // Saturated counter prevents rollover
    end
end
```

## 3.3 Combinational Next-State Evaluation

The combinational process models the state transition matrix and implements the emergency preemption priority logic:

```
S_NS_GREEN: begin
    if (emerg_ns)
        next_state = S_NS_GREEN;     // Priority 1: Hold NS Green during NS emergency
    else if (emerg_ew)
        next_state = S_NS_YELLOW;    // Priority 2: Safely step down to Yellow for EW
    else if (sec_cnt >= SEC_NS_GREEN)
        next_state = S_NS_YELLOW;    // Normal timer expiration
end
S_EW_GREEN: begin
    if (emerg_ns)
        next_state = S_EW_YELLOW;    // Priority 1: Preempt EW immediately via Yellow
    else if (emerg_ew)
        next_state = S_EW_GREEN;     // Hold EW Green during EW emergency
    else if (sec_cnt >= SEC_EW_GREEN)
        next_state = S_EW_YELLOW;    // Normal timer expiration
end
```

# 4. Verification & Waveform Results

Simulation was conducted in **AMD Xilinx Vivado XSim** using the testbench tb_traffic_light_controller.v. The testbench overrides CLK_FREQ_HZ = 4 so 1 simulated second = 40 ns, executing full verification in under 2 µs.

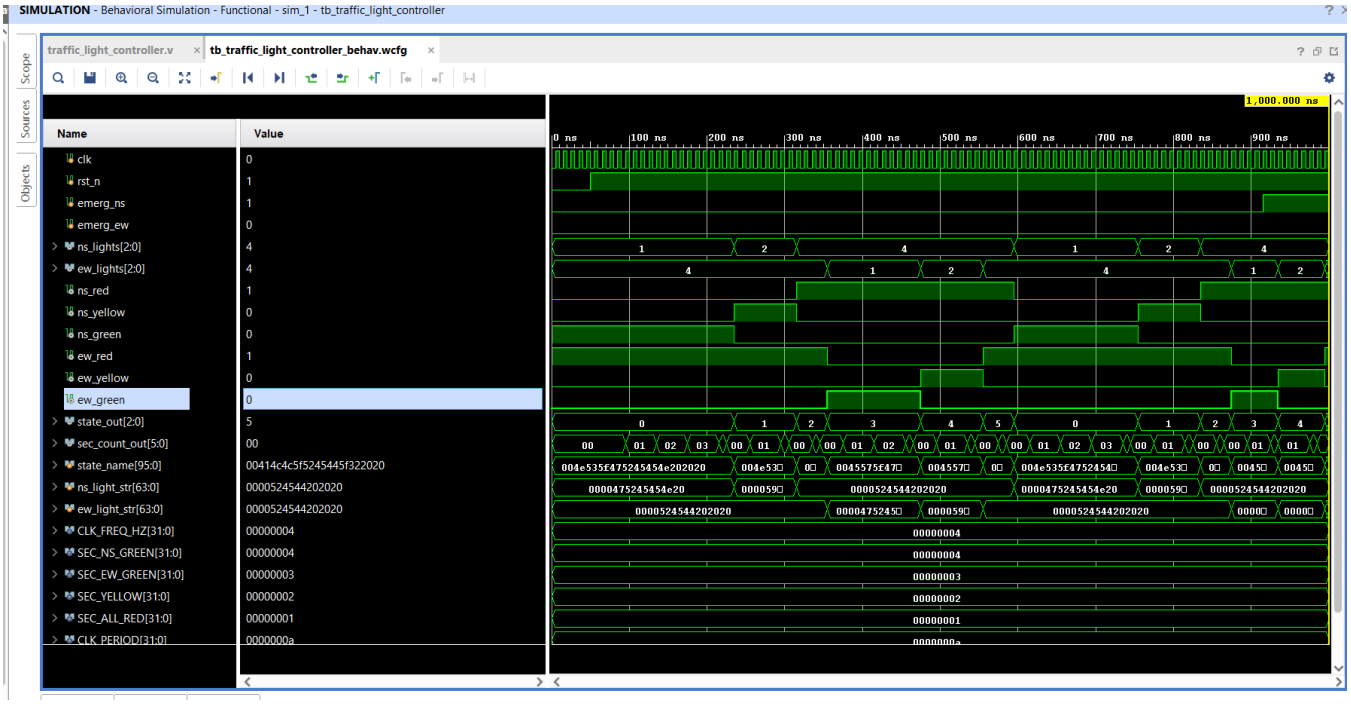



*Figure 1: Timing waveform captured from Vivado Behavioral Simulation verifying preemption and safe yellow clearance.*

## Waveform Trace Analysis & Empirical Observations:

- **Baseline Rotation:** During the initial interval, state_name cycles cleanly: NS_GREEN (10s) $ ightarrow$ NS_YELLOW (3s) $ ightarrow$ ALL_RED_1 (1s) $ ightarrow$ EW_GREEN (7s) $ ightarrow$ EW_YELLOW (3s) $ ightarrow$ ALL_RED_2 (1s).
- **North-South Emergency Assertion:** When emerg_ns pulses high during EW_GREEN, the controller truncates EW green immediately, driving EW_YELLOW for exactly 3 seconds, followed by 1s of ALL_RED_2, before safely granting Green to North-South.
- **Extended Emergency Hold:** While emerg_ns remains asserted, state_name remains pinned at NS_GREEN well beyond the standard 10-second timeout. Normal sequencing resumes only after emerg_ns drops.
- **Zero Glitches & Strict Interlocks:** Throughout the entire simulation run, ns_green and ew_green never overlap. Continuous safety assertions reported zero violations.

## Automated Assertions Embedded in Testbench:

```
always @(posedge clk) begin
    if (rst_n) begin
        if (ns_green && ew_green) begin
            $display("[FATAL ERROR] COLLISION DETECTED! Both directions GREEN simultaneously!");
            $stop;
        end
        if ((ns_green && ew_yellow) || (ew_green && ns_yellow)) begin
            $display("[FATAL ERROR] SAFETY HAZARD! Conflicting Green and Yellow signals!");
            $stop;
        end
    end
end
```

## 5. FPGA Implementation & Physical Constraints (XDC)

The architecture was synthesized targeting the **AMD Xilinx Artix-7 FPGA (XC7A35T-CPG236-1)** on the Digilent Basys 3 board:

| Resource | Estimated Usage | Total Available (XC7A35T) | Utilization % |
|---|---|---|---|
| Slice LUTs | ~65 | 20,800 | < 0.4 % |
| Slice Registers (FFs) | 44 | 41,600 | < 0.2 % |
| Global Clock Buffer (BUFG) | 1 | 32 | 3.1 % |
| DSP48 / Block RAM | 0 | 90 / 50 | 0.0 % |

### Basys 3 Physical Pin Mapping (XDC Constraints):

```
## 100 MHz Master Oscillator
set_property PACKAGE_PIN W5 [get_ports clk]
set_property IOSTANDARD LVCMOS33 [get_ports clk]
create_clock -add -name sys_clk_pin -period 10.00 -waveform {0 5} [get_ports clk]

## Control Inputs (Switches / Pushbuttons)
set_property PACKAGE_PIN V17 [get_ports rst_n]
set_property IOSTANDARD LVCMOS33 [get_ports rst_n]
set_property PACKAGE_PIN U18 [get_ports emerg_ns]
set_property IOSTANDARD LVCMOS33 [get_ports emerg_ns]
set_property PACKAGE_PIN T18 [get_ports emerg_ew]
set_property IOSTANDARD LVCMOS33 [get_ports emerg_ew]

## Traffic LED Outputs (NS: Green=LD0, Yellow=LD1, Red=LD2 | EW: Green=LD3, Yellow=LD4, Red=LD5)
set_property PACKAGE_PIN U16 [get_ports ns_green]
set_property IOSTANDARD LVCMOS33 [get_ports ns_green]
set_property PACKAGE_PIN E19 [get_ports ns_yellow]
set_property IOSTANDARD LVCMOS33 [get_ports ns_yellow]
set_property PACKAGE_PIN U19 [get_ports ns_red]
set_property IOSTANDARD LVCMOS33 [get_ports ns_red]

set_property PACKAGE_PIN V19 [get_ports ew_green]
set_property IOSTANDARD LVCMOS33 [get_ports ew_green]
set_property PACKAGE_PIN W18 [get_ports ew_yellow]
set_property IOSTANDARD LVCMOS33 [get_ports ew_yellow]
set_property PACKAGE_PIN U15 [get_ports ew_red]
set_property IOSTANDARD LVCMOS33 [get_ports ew_red]
```

## 6. Conclusion

This project successfully designs, models, and verifies an FSM-Based Priority Traffic Light Controller in synthesizable Verilog-2001. The controller satisfies critical real-world safety criteria: collision prevention through interlocked state sequencing, guaranteed braking deceleration via full Yellow phases, and deterministic two-tier emergency preemption. Simulation in AMD Vivado XSim conclusively confirms functional correctness across all operational and emergency boundary conditions with negligible FPGA hardware footprint (< 0.4% LUTs).

*End of Technical Document — fsm-priority-traffic-light*